

Direct Adaptive Cooperative Output Regulation of Unknown Multiagent Systems via Distributed Internal Model

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Abstract—The existing result on the cooperative output regulation problem for unknown linear multiagent systems using the data-driven distributed internal model approach is limited to the case where each follower is a single-input and single-output system, and the communication network among all agents is an acyclic static digraph. In this article, we further address the same problem for unknown linear multiagent systems with multi-input and multi-output followers over a general static and connected digraph by a value iteration approach. Further, we make two main improvements over the existing result. First, compared with the existing approach, we reduce the number of the unknown variables governed by a sequence of linear algebraic equations. Second, we show that the sequence of linear algebraic equations can be further decoupled to two sequences of lower dimensional linear algebraic equations. As a result, our approach not only drastically reduces the computational cost but also significantly weakens the solvability conditions. A numerical example is used to illustrate the effectiveness of our approach.

Index Terms—Cooperative output regulation, distributed internal model, multiagent systems (MASs), reinforcement learning (RL).

I. INTRODUCTION

The cooperative output regulation problem (CORP) is a generalization of the output regulation problem (ORP) for a single-agent system to a multiagent system (MAS) [1], [2], [3]. There are two approaches for dealing with the ORP: feedforward design and internal model principle [4], [5], [6], [7]. Extending these two approaches to MASs leads to the so-called distributed-observer approach [1], [3] and distributed-internal-model approach [2], [3]. A drawback of the feedforward design and hence the distributed observer approach is that they both make use of the feedforward control, which relies on the solution of the regulator equations, and thus requires the exact system model.

Recently, there is a surge of interest in studying both ORP for unknown linear systems and CORP for unknown linear MASs, thus leading to the so-called data-driven approach, which integrates the reinforcement learning (RL) [8], [9], also called the adaptive dynamic programming (ADP), technique and the feedforward design or internal model principle for dealing with ORP, or integrates the RL technique and the distributed-observer approach or the distributed-internal-model approach to deal with CORP.

The RL technique has broad applications in many areas. Its application to the linear quadratic regulator (LQR) problem for unknown

continuous-time linear systems can be traced back to [10], which studied the LQR problem for linear systems with unknown system matrix by a policy-iteration (PI) method. Jiang and Jiang [11] further extended the result of [10] to linear systems with unknown system matrix and input matrix by the PI-based method. However, the PI-based method requires an initially stabilizing feedback gain to start the iteration process. Bian and Jiang [12] broke this barrier by proposing a value-iteration (VI) method to tackle the same problem as that in [11], which does not need an initially stabilizing feedback gain to start the algorithm. On the basis of the approaches in [11] and [12], the authors in [13] and [14] studied the optimal ORP for unknown linear systems by combining the feedforward design with the PI-based method and VI-based method, respectively. Lin and Huang [15] further considered the optimal ORP for unknown linear systems by combining the internal model principle with the VI-based method.

By integrating the technique presented in [13] and the distributed observer detailed in [1], Gao et al. [16] considered the optimal CORP for linear MAS with unknown state equations. Lin and Huang [17] improved the approach in [16] by reducing the computational cost and weakening the solvability conditions. Due to the use of feedforward control, the approaches in [16] and [17] require the output equation to be known. To deal with the unknown output equation, Gao et al. [18] developed both PI-based and VI-based methods to handle the optimal CORP by combining the distributed-observer and distributed-internal-model approach.

However, the approach in [18] only applies to MAS with single-input and single-output (SISO) followers over acyclic communication networks. In this article, we will further consider the CORP for MAS with multi-input and multi-output (MIMO) followers without assuming that the network is acyclic. Moreover, all the data-driven approaches to ORP or CORP of linear systems boil down to solving a sequence of linear algebraic equations governing a set of unknown variables. Thus, the computational cost and the solvability conditions depend on the complexity of these equations. For this reason, we will further make two main improvements over the existing results. First, due to the use of the virtual tracking error, the closed-loop systems of various followers in [18] are coupled. Thus, the learning process in [18] is quite complicated, and moreover, it yields a set of high dimensional linear algebraic equations. In contrast, by normalizing the virtual error and making some additional manipulations, we decouple the closed-loop systems of various followers, and thus lead to a set of linear algebraic equations with much lower dimension. We further show that the unknown variables of these linear equations can be decomposed into two groups and thus the original derived sequence of linear equations can be reduced to two sets of lower dimensional linear algebraic equations. As a result of these two improvements, our approach not only drastically reduces the computational cost but also significantly weakens the solvability conditions in terms of the full column rank requirements for these equations.

The rest of this article is organized as follows. Section II provides preliminaries for the CORP based on the distributed internal model

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method. Section III presents a novel VI-based approach for solving the CORP. Section IV gives a numerical example to illustrate the effectiveness of our proposed approach. Finally, Section V concludes this article with some concluding remarks.

Notations: Throughout this article, $\mathbb{R}, \mathbb{N}, \mathbb{N}_+$, and $\mathbb{C}_-$ represent the sets of real numbers, nonnegative integers, positive integers, and the open left-half complex plane, respectively. $\mathbb{S}^n$ is the $n \times n$ real symmetric matrix space. $\mathcal{P}^n$ is the set of all $n \times n$ real, symmetric, and positive semidefinite matrices. $|\cdot|$ represents the Euclidean norm for vectors and the induced norm for matrices. $\otimes$ denotes the Kronecker product. For $b = [b_1, b_2, \dots, b_n]^T \in \mathbb{R}^n$, $\text{vecv}(b) = [b_1^2, b_1 b_2, \dots, b_1 b_n, b_2^2, b_2 b_3, \dots, b_{n-1} b_n, b_n^2]^T \in \mathbb{R}^{\frac{n(n+1)}{2}}$. For a symmetric matrix $P = [p_{ij}]_{n \times n} \in \mathbb{R}^{n \times n}$, $\text{vecs}(P) = [p_{11}, 2p_{12}, \dots, 2p_{1n}, p_{22}, 2p_{23}, \dots, 2p_{n-1,n}, p_{nn}]^T \in \mathbb{R}^{\frac{n(n+1)}{2}}$. For $v \in \mathbb{R}^n$, $|v|_P = v^T P v$. For column vectors $a_i, i = 1, \dots, s$, $\text{col}(a_1, \dots, a_s) = [a_1^T, \dots, a_s^T]^T$, and if $A = (a_1, \dots, a_s)$, then $\text{vec}(A) = \text{col}(a_1, \dots, a_s)$. For $A \in \mathbb{R}^{n \times n}$, $\sigma(A)$ denotes the spectrum of A . “blockdiag” denotes the block diagonal matrix operator. I_n denotes the identity matrix of dimension n .

II. PRELIMINARY

In this section, we review some results on the CORP via the distributed internal model approach based on [2], [3].

Consider a class of linear MASs in the following form:

$$\begin{aligned}\dot{x}_i &= Ax_i + Bu_i + E_i v \\ y_i &= Cx_i\end{aligned}\quad (1)$$

where, for $i = 1, 2, \dots, N$, $x_i \in \mathbb{R}^n$, $y_i \in \mathbb{R}^p$, and $u_i \in \mathbb{R}^m$ are the state, measurement output, and control input of the i th agent, respectively, and $v \in \mathbb{R}^q$ is the state of the exosystem as follows:

$$\dot{v} = Sv. \quad (2)$$

Define the tracking error of the i th subsystem as follows:

$$e_i = y_i - y_0 = Cx_i + Fv \quad (3)$$

where $y_0 = -Fv$ is the reference signal.

Like in [2], we treat the exosystem (2) and the plant (1) as a leader–follower MAS with $N + 1$ agents where (2) is the leader and the N subsystems of (1) are N followers. To describe the communication constraints of this leader–follower system, we define a digraph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ where $\mathcal{V} = \{0, 1, \dots, N\}$ is the node set and $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ is the edge set. The node 0 is associated with the leader (2) and the remaining N nodes are associated with the N followers of (1). For $i, j = 0, 1, \dots, N$, $i \neq j$, an edge from node j to node i is denoted by (j, i) , where the nodes j and i are called the parent node and the child node of each other. A directed spanning tree is a digraph in which every node has exactly one parent except for one node, called the root, which has no parent and from which every other node is reachable. Let $\mathcal{N}_i = \{j | (j, i) \in \mathcal{E}\}$, which is called the neighbor set of node i , and let $\mathcal{N}_i^- = \mathcal{N}_i \setminus \{0\}$. Let $A = [a_{ij}] \in \mathbb{R}^{(N+1) \times (N+1)}$ be such that $a_{ij} = 1$, if $(j, i) \in \mathcal{E}$; otherwise $a_{ij} = 0$, which is called the adjacency matrix of $\mathcal{G}$. We assume that $a_{0j} = 0$ for $j = 1, 2, \dots, N$ since there are no edges from any follower to the leader. Let $H = [h_{ij}] \in \mathbb{R}^{N \times N}$ where, for $i, j = 1, 2, \dots, N$, $h_{ii} = \sum_{j=0}^N a_{ij}$ and $h_{ij} = -a_{ij}$ for all $i \neq j$.

Like in [2], we describe the CORP as follows.

Problem 1: Design a distributed controller u_i such that the closed-loop system is exponentially stable with v set to zero and, for $i = 1, \dots, N$, $\lim_{t \rightarrow \infty} e_i(t) = 0$.

Some standard assumptions needed for solving Problem 1 are listed as follows.

Assumption 1: The pair (A, B) is stabilizable.

Assumption 2: All the eigenvalues of S have nonnegative real parts.

Assumption 3: $\text{rank} \begin{pmatrix} A - \lambda I & B \\ C & 0 \end{pmatrix} = n + p, \forall \lambda \in \sigma(S)$.

Assumption 4: The digraph $\mathcal{G}$ contains a directed spanning tree with the node 0 as its root.

In order to make use of the internal model principle, we present the concept of minimal p – copy internal model of the matrix S as follows.

Definition 1: A pair of matrices (G_1, G_2) is said to incorporate the minimum p – copy internal model of the matrix S if

$$G_1 = \text{blockdiag} \underbrace{[\beta, \dots, \beta]}_{p\text{-tuple}}, \quad G_2 = \text{blockdiag} \underbrace{[\sigma, \dots, \sigma]}_{p\text{-tuple}} \quad (4)$$

where β is a constant square matrix whose characteristic polynomial equals the minimal polynomial of S , and σ is a constant column vector such that (β, σ) is controllable.

To describe our distributed control law, let the virtual tracking error e_{vi} for the i th subsystem be as follows [2]

$$e_{vi} = \sum_{j \in \mathcal{N}_i^-} a_{ij}(y_i - y_j). \quad (5)$$

Then, the distributed dynamic state feedback control law proposed in [2] is as follows:

$$\begin{aligned}u_i &= K_x \left(\sum_{j \in \mathcal{N}_i^-} a_{ij}(x_i - x_j) + a_{i0}x_i \right) + K_z z_i \\ \dot{z}_i &= G_1 z_i + G_2 e_{vi}\end{aligned}\quad (6)$$

where K_x and K_z are some constant matrices to be designed, and $z_i \in \mathbb{R}^{n_z}$ for some integer n_z .

The solvability of Problem 1 was first studied in [2] and further refined in [3]. The following result taken from [2] summarizes the solvability of Problem 1.

Lemma 1: Under Assumptions 1–4, there exists a control gain (K_x, K_z) such that the following matrix:

$$A_c \triangleq \begin{bmatrix} I_N \otimes A + H \otimes (BK_x) & I_N \otimes (BK_z) \\ H \otimes (G_2 C) & I_N \otimes G_1 \end{bmatrix} \quad (7)$$

is Hurwitz, and moreover, any control gain (K_x, K_z) that makes A_c Hurwitz is such that (6) solves Problem 1.

Remark 1: Let $Y \triangleq \begin{bmatrix} A & 0 \\ G_2 C & G_1 \end{bmatrix}$ and $J \triangleq \begin{bmatrix} B \\ 0 \end{bmatrix}$. Under Assumptions 1–3, by [7, Lemma 1.26], (Y, J) is stabilizable. Thus, the Riccati Equation

$$Y^T P + PY - PJJ^T P + I_{n+n_z} = 0 \quad (8)$$

admits a unique positive definite solution P^* . By [3, Lemma 9.3], any control gain

$$[K_x, K_z] \triangleq K = -\nu^{-1} J^T P^* \quad (9)$$

with $0 < \nu \leq 2\text{Re}(\lambda_i), \forall \lambda_i \in \sigma(H), i = 1, \dots, N$, makes A_c Hurwitz.

We now summarize the main result of [2] as follows.

Theorem 1: Under Assumptions 1–4, let (G_1, G_2) incorporate a minimum p – copy internal model of the matrix S . Then, Problem 1 is solved by the distributed dynamic feedback control law (6) with the feedback control gain (9).

Remark 2: Two significant differences between the method of [18] and the method here are worth mentioning. First, the internal model used in [18] is 1-copy, which is a special case of what is used here with

$G_2 = [0, 0, \dots, 0, 1]^T$ be a column vector. Thus, every subsystem of the follower system in [18] must be SISO. In contrast, each subsystem of the follower system here can be MIMO since we use the p-copy internal model. Second, the approach of [18] only applies to the case where the closed-loop compact system matrix A_c in [18, eq. (16)] is lower triangular, which is possible only if the communication graph is acyclic. In contrast, since we resort to the technique of [3, Lemma 9.3], which is able to place the eigenvalues of N systems simultaneously, we do not require the closed-loop compact system matrix A_c to be a lower triangular one, and thus our Assumption 4 here accommodates graphs with cycles.

III. ADAPTIVE COOPERATIVE OUTPUT REGULATION

In this section, we further consider solving Problem 1 without knowing the matrices A, B, C, E_i , and F by a data-driven approach. The key for solving this problem is to obtain the solution of the Riccati equation (8) without knowing the matrices A, B, C, E_i , and F . For this purpose, we need to relate the system's dynamics to the Riccati equation to be solved. Then, with the virtual error e_{vi} given by (5), we can obtain the following so-called augmented system:

$$\begin{aligned}\dot{x}_i &= Ax_i + Bu_i + E_i v \\ \dot{z}_i &= G_1 z_i + G_2 \sum_{j \in \mathcal{N}_i} a_{ij} (y_i - y_j).\end{aligned}\quad (10)$$

Let $\xi_i \triangleq \text{col}(x_i, z_i)$. Then, ξ_i is governed by the following differential equation:

$$\dot{\xi}_i = \hat{Y} \xi_i + Ju_i + \begin{bmatrix} E_i \\ 0 \end{bmatrix} v + \gamma_i \quad (11)$$

where $\hat{Y} = \begin{bmatrix} A & 0 \\ (\sum_{j \in \mathcal{N}_i} a_{ij}) G_2 C & G_1 \end{bmatrix}$ and $\gamma_i = -\begin{bmatrix} 0 \\ \sum_{j \in \mathcal{N}_i} (a_{ij} G_2 y_j) \end{bmatrix}$. Since $\hat{Y} \neq Y$, we cannot relate (11) to the Riccati equation (8). To fix this problem, let

$$\hat{e}_{vi} = \frac{e_{vi}}{\sum_{j \in \mathcal{N}_i} a_{ij}} \quad (12)$$

which is the normalized version of e_{vi} . Then, we obtain the following modified augmented system:

$$\begin{aligned}\dot{x}_i &= Ax_i + Bu_i + E_i v \\ \dot{\hat{z}}_i &= G_1 \hat{z}_i + G_2 \hat{e}_{vi}\end{aligned}\quad (13)$$

and the modified distributed control law

$$u_i = \frac{K_x}{\sum_{j \in \mathcal{N}_i} a_{ij}} \left(\sum_{j \in \mathcal{N}_i} a_{ij} (x_i - x_j) + a_{i0} x_i \right) + K_z \hat{z}_i. \quad (14)$$

Let $x = \text{col}(x_1, \dots, x_N)$, $u = \text{col}(u_1, \dots, u_N)$, $\hat{e}_v = \text{col}(\hat{e}_{v1}, \dots, \hat{e}_{vN})$, $\hat{z} = \text{col}(\hat{z}_1, \dots, \hat{z}_N)$, $\tilde{A} = I_N \otimes A$, $\tilde{B} = I_N \otimes B$, $\tilde{E} = \text{col}(E_1, \dots, E_N)$, $\tilde{G}_1 = I_N \otimes G_1$, $\tilde{G}_2 = I_N \otimes G_2$, $\tilde{C} = (DH) \otimes C$, $\tilde{F} = (DH \mathbf{1}_N) \otimes F$, and $D = \text{blockdiag} \left(\frac{1}{\sum_{j \in \mathcal{N}_1} a_{1j}}, \frac{1}{\sum_{j \in \mathcal{N}_2} a_{2j}}, \dots, \frac{1}{\sum_{j \in \mathcal{N}_N} a_{Nj}} \right)$. Then, the compact form of (13) is given by

$$\begin{aligned}\dot{x} &= \tilde{A}x + \tilde{B}u + \tilde{E}v \\ \dot{\hat{z}} &= \tilde{G}_1 \hat{z} + \tilde{G}_2 \hat{e}_v \\ \hat{e}_v &= \tilde{C}x + \tilde{F}v.\end{aligned}\quad (15)$$

We have the following result.

Lemma 2: Under Assumptions 1–4, the CORP of (15) is solved by the distributed controller (14) if (K_x, K_z) is such that the following matrix $\hat{A}_c$ is Hurwitz:

$$\hat{A}_c \triangleq \begin{bmatrix} I_N \otimes A + (DH) \otimes (BK_x) & I_N \otimes (BK_z) \\ (DH) \otimes (G_2 C) & I_N \otimes G_1 \end{bmatrix}. \quad (16)$$

Proof: The compact form of (14) is as follows:

$$u = ((DH) \otimes K_x)x + (I_N \otimes K_z)\hat{z}. \quad (17)$$

Substituting (17) into (15) gives the following closed-loop system:

$$\begin{aligned}\dot{x} &= (\tilde{A} + \tilde{B}((DH) \otimes K_x))x + \tilde{B}(I_N \otimes K_z)\hat{z} + \tilde{E}v \\ \dot{\hat{z}} &= \tilde{G}_2 \tilde{C}x + \tilde{G}_1 \hat{z} + \tilde{G}_2 \tilde{F}v \\ \hat{e}_v &= \tilde{C}x + \tilde{F}v.\end{aligned}\quad (18)$$

Let

$$\hat{A}_c = \begin{bmatrix} \tilde{A} + \tilde{B}((DH) \otimes K_x) & \tilde{B}(I_N \otimes K_z) \\ \tilde{G}_2 \tilde{C} & \tilde{G}_1 \end{bmatrix} \quad (19)$$

which is the system matrix of the closed-loop system (18) and is the same as the matrix on the right-hand side of (16). Suppose (K_x, K_z) is such that $\hat{A}_c$ is Hurwitz. Since $(\tilde{G}_1, \tilde{G}_2)$ incorporates a minimum Np -copy internal model of the matrix S , by [7, Lemma 1.27], for any $\tilde{E}$ and $\tilde{F}$, the following matrix equations have a unique solution (X, Z) :

$$\begin{aligned}XS &= \tilde{A}X + \tilde{B}[(DH) \otimes K_x]X + (I_N \otimes K_z)Z + \tilde{E} \\ ZS &= \tilde{G}_1 Z + \tilde{G}_2 (\tilde{C}X + \tilde{F}) \\ 0 &= \tilde{C}X + \tilde{F}.\end{aligned}\quad (20)$$

Let $\beta_x = x - Xv$ and $\beta_z = \hat{z} - Zv$. Then, it can be verified that

$$\begin{bmatrix} \dot{\beta}_x \\ \dot{\beta}_z \end{bmatrix} = \hat{A}_c \begin{bmatrix} \beta_x \\ \beta_z \end{bmatrix}, \quad \hat{e}_v = \tilde{C}\beta_x.$$

Since $\hat{A}_c$ is Hurwitz, $\lim_{t \rightarrow \infty} \hat{e}_v = 0$. Note that $\hat{e}_v = (DH \otimes I_p)e$ with $e = \text{col}(e_1, \dots, e_N)$. It can be seen that, under Assumption 4, D is nonsingular. Also, by [19, Lemma 4], under Assumption 4, all the eigenvalues of the matrix H have positive real parts. Thus, we also have $\lim_{t \rightarrow \infty} e = 0$.

We are now ready to present the following result.

Theorem 2: Under Assumptions 1–4, Problem 1 is solved by the distributed dynamic feedback control law (14) with the following feedback control gain:

$$[K_x, K_z] \triangleq K = -\omega^{-1} J^T P^* \quad (21)$$

where P^* is the solution to the Riccati equation (8) and $0 < \omega \leq 2\text{Re}(\lambda_i), \forall \lambda_i \in \sigma(DH), i = 1, \dots, N$.

Proof: The theorem has been established for the special case where D is an identity matrix in [3, Lemma 9.3]. Since the only property of the matrix H used in the proof of [3, Lemma 9.3] is that all the eigenvalues of the matrix H have positive real parts, which, under Assumption 4, is guaranteed by [19, Lemma 4]. Since, under Assumption 4, D is a diagonal matrix with all diagonal elements positive, all the eigenvalues of the matrix DH have positive real parts. Thus, the proof of Theorem 2

is the same as that of [3, Lemma 9.3] upon replacing H by DH everywhere.

Let $\hat{\xi}_i \triangleq \text{col}(x_i, \hat{z}_i)$. Then, under the control law (14), $\hat{\xi}_i$ is governed by the following differential equation:

$$\dot{\hat{\xi}}_i = Y\hat{\xi}_i + Ju_i + \begin{bmatrix} E_i \\ 0 \end{bmatrix} v + \hat{\gamma}_i \quad (22)$$

$$\text{where } \hat{\gamma}_i = - \begin{bmatrix} 0 \\ \frac{\sum_{j \in \mathcal{N}_i} (a_{ij} G_2 y_j)}{\sum_{j \in \mathcal{N}_i} a_{ij}} \end{bmatrix}.$$

Based on (22), we can apply the off-policy RL/ADP method to approximately solve the Riccati equation (8) and obtain the controller without the information of the system matrices provided that the data of $\hat{\xi}_i, u_i, v$, and $\hat{\gamma}_i$ are available. However, due to the communication constraints, some followers cannot directly access the information of the exosystem's state v , i.e., agent i can access the signal v if and only if $0 \in \mathcal{N}_i$. To circumvent this difficulty, we will introduce the following distributed observer proposed in [1] to estimate v :

$$\dot{\eta}_i = S\eta_i + \mu \sum_{j \in \mathcal{N}_i} a_{ij}(\eta_j - \eta_i), \quad i = 1, \dots, N \quad (23)$$

where $\eta_i \in \mathbb{R}^q$ for $i = 0, 1, \dots, N$ with $\eta_0 \triangleq v$, and $\mu > \frac{\lambda_{\max}^{\text{re}}(S)}{\lambda_{\min}^{\text{re}}(H)}$, where $\lambda_{\max}^{\text{re}}(\cdot)$ and $\lambda_{\min}^{\text{re}}(\cdot)$ represent the largest and smallest real parts of the eigenvalues of a matrix, respectively. By [1, Thm. 1], $(\eta_i - v)$ tends to 0 exponentially.

Remark 3: Instead of (23), the authors of [18] made use of the following adaptive distributed observer proposed in [20] to estimate the state v of the exosystem:

$$\begin{aligned} \dot{S}_i &= \mu_1 \sum_{j \in \mathcal{N}_i} a_{ij}(S_j - S_i) \\ \dot{\eta}_i &= S_i \eta_i + \mu_2 \sum_{j \in \mathcal{N}_i} a_{ij}(\eta_j - \eta_i), \quad i = 1, \dots, N \end{aligned} \quad (24)$$

where $S_0 \triangleq S$ and $\mu_1, \mu_2 > 0$ are some positive constants. It was shown in [20] that there exist two real positive numbers μ_{10} and μ_{20} determined by the eigenvalues of S and H such that, for all $\mu_1 > \mu_{10}$ and $\mu_2 > \mu_{20}$, the solutions of (24) are such that, for $i = 1, \dots, N$, S_i tend to S and η_i tend to v , exponentially. Thus, an advantage of (24) over (23) is that it only requires the children of the leader to be able to access the matrix S . Nevertheless, this advantage is paid by more computational cost of (24) over (23). In this article, for better elucidating the advantage of our approach, we choose to use (23) instead of (24).

Now, rewrite (22) to the following form:

$$\dot{\hat{\xi}}_i = Y\hat{\xi}_i + Ju_i + \begin{bmatrix} E_i \\ 0 \end{bmatrix} \eta_i + \rho_i + \hat{\gamma}_i \quad (25)$$

where $\rho_i = \begin{bmatrix} E_i \\ 0 \end{bmatrix} (v - \eta_i)$ tends to 0 exponentially.

Remark 4: Two points are worth mentioning. First, $\hat{\gamma}_i$ satisfies the communication constraints and is thus accessible by agent i through the communication network. Second, since ρ_i tends to 0 exponentially, we can neglect the term ρ_i in (25) by starting the data collecting process after some finite time $t_0 > 0$. Specifically, let $\frac{1}{\tau} < 0$ be the largest real part of the eigenvalues of the matrix $(I_N \otimes S) - \mu(H \otimes I_q)$. Then, we

Algorithm 1: VI Algorithm for Solving (8) [12].

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Choose  $P_0 = P_0^T > 0, k, q \leftarrow 0$ .
loop
   $\tilde{P}_{k+1} \leftarrow P_k + \epsilon_k(Y^T P_k + P_k Y - P_k J J^T P_k + I_{n+n_z})$ 
  if  $\tilde{P}_{k+1} \notin B_q$  then
     $P_{k+1} \leftarrow P_0, q \leftarrow q + 1$ .
  else if  $|\tilde{P}_{k+1} - P_k|/\epsilon_k < \varepsilon$  then
    return  $P_k$  as an approximation to  $P^*$ 
  else
     $P_{k+1} \leftarrow \tilde{P}_{k+1}$ 
  end if
   $k \leftarrow k + 1$ 
end loop

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can set $t_0 \geq 5\tau$. It is noted that τ can be made arbitrarily small by having μ sufficiently large.

By ignoring the term ρ_i for $t \geq t_0$, (25) can be simplified to the following form:

$$\dot{\hat{\xi}}_i = Y\hat{\xi}_i + Ju_i + \begin{bmatrix} E_i \\ 0 \end{bmatrix} \eta_i + \hat{\gamma}_i, \quad t \geq t_0. \quad (26)$$

In what follows, we will develop a VI-based approach to solve Problem 1 based on (26).

A. VI-Based Approach

Gao et al. [18] developed a VI-based algorithm for data-driven CORP of MASs with SISO followers over acyclic graphs based on the algorithm of [12]. In this section, we further deal with the more general case where all the followers can be MIMO and the graph does not have to be acyclic.

Like in [12], let ϵ_k be a series of time steps satisfying

$$\epsilon_k > 0, \sum_{k=0}^{\infty} \epsilon_k = \infty, \sum_{k=0}^{\infty} \epsilon_k^2 < \infty. \quad (27)$$

$\{B_q\}_{q=0}^{\infty}$ is a collection of bounded sets satisfying

$$B_q \subset B_{q+1}, \quad q \in \mathbb{N}, \quad \lim_{q \rightarrow \infty} B_q = \mathcal{P}^{n+n_z} \quad (28)$$

and $\varepsilon > 0$ is the convergence criterion. Then, by applying the algorithm from [12], we obtain Algorithm 1 to approximate the solution to (8).

By [12, Thm 3.3], Algorithm 1 is such that $\lim_{k \rightarrow \infty} P_k = P^*$.

For simplicity, define $H_k = Y^T P_k + P_k Y, K_k = -J^T P_k$. From (26), we have

$$\begin{aligned} & |\hat{\xi}_i(t + \delta t)|_{P_k} - |\hat{\xi}_i(t)|_{P_k} \\ &= \int_t^{t+\delta t} \left[|\hat{\xi}_i|_{H_k} - 2u_i^T K_k \hat{\xi}_i + 2\eta_i^T \begin{bmatrix} E_i \\ 0 \end{bmatrix}^T P_k \hat{\xi}_i \right. \\ & \quad \left. + 2\hat{\gamma}_i^T P_k \hat{\xi}_i \right] d\tau. \end{aligned} \quad (29)$$

For convenience, define a constant matrix $M \in \mathbb{R}^{(n+n_z)^2 \times \frac{(n+n_z)(n+n_z+1)}{2}}$ such that $M \text{vecs}(Q) = \text{vec}(Q)$ for any symmetric matrix $Q \in \mathbb{R}^{(n+n_z) \times (n+n_z)}$.

Furthermore, for any vectors $a \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, and any integer $s \in \mathbb{N}_+$, define

$$\begin{aligned} \delta_a &= [\text{vecv}(a(t_1)) - \text{vecv}(a(t_0)), \dots, \\ &\quad \text{vecv}(a(t_s)) - \text{vecv}(a(t_{s-1}))]^T \\ \Gamma_{ab} &= \left[\int_{t_0}^{t_1} a \otimes b d\tau, \int_{t_1}^{t_2} a \otimes b d\tau, \dots, \int_{t_{s-1}}^{t_s} a \otimes b d\tau \right]^T. \end{aligned} \quad (30)$$

Then, (29) together with (30) gives

$$\hat{\Psi}_i \begin{bmatrix} \text{vecs}(H_k) \\ \text{vec}(K_k) \\ \text{vec} \left(\begin{bmatrix} E_i \\ 0 \end{bmatrix}^T P_k \right) \end{bmatrix} = \hat{\Phi}_{ik} \quad (31)$$

where $\hat{\Psi}_i = [\hat{\Psi}_i^1, \hat{\Psi}_i^2, \hat{\Psi}_i^3]$, $\hat{\Psi}_i^1 = \Gamma_{\hat{\xi}_i \hat{\xi}_i} M$, $\hat{\Psi}_i^2 = -2\Gamma_{\hat{\xi}_i u_i} \hat{\Psi}_i^3 = 2\Gamma_{\hat{\xi}_i \eta_i}$, and $\hat{\Phi}_{ik} = [\delta_{\hat{\xi}_i} - 2\Gamma_{\hat{\xi}_i \hat{\gamma}_i} M] \text{vecs}(P_k)$.

The solvability of (31) is guaranteed by the following lemma.

Lemma 3: The matrix $\hat{\Psi}_i$ in (31) has full column rank if

$$\begin{aligned} \text{rank}([\Gamma_{\hat{\xi}_i \hat{\xi}_i}, \Gamma_{\hat{\xi}_i u_i}, \Gamma_{\hat{\xi}_i \eta_i}]) &= \frac{(n+n_z)(n+n_z+1)}{2} \\ &\quad + (n+n_z)(m+q). \end{aligned} \quad (32)$$

Proof: Given any column vector $a \in \mathbb{R}^{n+n_z}$ and any symmetric matrix $Q \in \mathbb{R}^{(n+n_z) \times (n+n_z)}$, it can be verified that $a^T Q a = (a \otimes a)^T \text{vec}(Q) = \text{vecv}(a)^T \text{vecs}(Q)$. Furthermore, recalling that $\text{vec}(Q) = M \text{vecs}(Q)$ from the definition of matrix M gives $(a \otimes a)^T M \text{vecs}(Q) = \text{vecv}(a)^T \text{vecs}(Q)$.

Since Q can be any symmetric matrix, we obtain $(a \otimes a)^T M = \text{vecv}(a)^T$ for any column vector $a \in \mathbb{R}^{n+n_z}$. Thus, $\Gamma_{\hat{\xi}_i \hat{\xi}_i} M = \hat{\Gamma}_{\hat{\xi}_i}$, where $\hat{\Gamma}_{\hat{\xi}_i} = [\int_{t_0}^{t_1} \text{vecv}(\hat{\xi}_i) d\tau, \int_{t_1}^{t_2} \text{vecv}(\hat{\xi}_i) d\tau, \dots, \int_{t_{s-1}}^{t_s} \text{vecv}(\hat{\xi}_i) d\tau]^T$.

Since

$$\hat{\Psi}_i = [\Gamma_{\hat{\xi}_i \hat{\xi}_i} M, -2\Gamma_{\hat{\xi}_i u_i}, 2\Gamma_{\hat{\xi}_i \eta_i}] = [\hat{\Gamma}_{\hat{\xi}_i}, \Gamma_{\hat{\xi}_i u_i}, \Gamma_{\hat{\xi}_i \eta_i}] \Theta$$

where $\Theta = \text{blockdiag}[\frac{(n+n_z)(n+n_z+1)}{2}, -2I_{m(n+n_z)}, 2I_{q(n+n_z)}]$, we have $\text{rank}(\hat{\Psi}_i) = \text{rank}([\hat{\Gamma}_{\hat{\xi}_i}, \Gamma_{\hat{\xi}_i u_i}, \Gamma_{\hat{\xi}_i \eta_i}])$. Since, by inspection, $\text{rank}([\hat{\Gamma}_{\hat{\xi}_i}, \Gamma_{\hat{\xi}_i u_i}, \Gamma_{\hat{\xi}_i \eta_i}]) = \text{rank}([\Gamma_{\hat{\xi}_i \hat{\xi}_i}, \Gamma_{\hat{\xi}_i u_i}, \Gamma_{\hat{\xi}_i \eta_i}])$, we have $\text{rank}(\hat{\Psi}_i) = \text{rank}([\Gamma_{\hat{\xi}_i \hat{\xi}_i}, \Gamma_{\hat{\xi}_i u_i}, \Gamma_{\hat{\xi}_i \eta_i}])$. In conclusion, (32) means $\hat{\Psi}_i$ has full column rank.

In Algorithm 1, $\tilde{P}_{k+1}$ can be expressed as $\tilde{P}_{k+1} \triangleq P_k + \epsilon_k (H_k - K_k^T K_k + I_{n+n_z})$. Thus, by iteratively applying the solution of (31) to Algorithm 1, one can obtain the approximate solution to (8) as well as the feedback control gain (21) without using the information of system matrices.

Remark 5: It is interesting to compare our approach with that of [18], which studied the CORP for SISO linear MASs over acyclic graph. The augmented system of [18] takes the following form:

$$\dot{\xi}_i = Y \xi_i + J u_i + \begin{bmatrix} E_i \\ \alpha_{i0} G_2 F \end{bmatrix} v - \sum_{j \in \mathcal{N}_i^-} \Theta_{ij} \xi_j \quad (33)$$

where ξ_i represents the augmented state of agent i in [18], $\Theta_{ij} = \begin{bmatrix} 0 & 0 \\ \alpha_{ij} G_2 C & 0 \end{bmatrix}$, and $\alpha_{ij} = \frac{a_{ij}}{\sum_{j \in \mathcal{N}_i^-} a_{ij}}$. Here, for fair comparison, we have ignored an error term in (33) caused by using the adaptive distributed observer (24) instead of the distributed observer (23), and set the weighting matrices Q and R in [18] to be identity matrices.

Based on (33), replace v with η_i and ignore the error caused by this replacement, the following equations are derived:

$$\Psi_i \begin{bmatrix} \text{vecs}(H_k) \\ \text{vec}(K_k) \\ \text{vec} \left(\begin{bmatrix} E_i \\ \alpha_{i0} G_2 F \end{bmatrix}^T P_k \right) \\ \text{vec}(\Xi_{ik}) \end{bmatrix} = \Phi_{ik} \quad (34)$$

where

$$\Psi_i = [\hat{\Gamma}_{\xi_i}, -2\Gamma_{\xi_i u_i}, 2\Gamma_{\xi_i \eta_i}, -2\Gamma_{\xi_i \psi_i}], \Phi_{ik} = \delta_{\xi_i} \text{vecs}(P_k)$$

$$\psi_i = [\xi_{j1}^T, \xi_{j2}^T, \dots, \xi_{j|\mathcal{N}_i^-|}^T]^T \in \mathbb{R}^{h_i}$$

$$\Xi_{ik} = [P_k \Theta_{ij1}, P_k \Theta_{ij2}, \dots, P_k \Theta_{ij|\mathcal{N}_i^-|}]^T.$$

The number of unknown variables in (34) is $\frac{(n+n_z)(n+n_z+1)}{2} + (n+n_z)(m+q+h_i)$, while our equation (31) only contains $\frac{(n+n_z)(n+n_z+1)}{2} + (n+n_z)(m+q)$ unknown variables. Thus, our method has reduced the number of the unknown variables by $(n+n_z)h_i$. Moreover, the solvability condition for (34) is

$$\begin{aligned} \text{rank}([\Gamma_{\xi_i \xi_i}, \Gamma_{\xi_i u_i}, \Gamma_{\xi_i \eta_i}, \Gamma_{\xi_i \psi_i}]) \\ = \frac{(n+n_z)(n+n_z+1)}{2} + (n+n_z)(m+q+h_i). \end{aligned} \quad (35)$$

Thus, checking (35) is much more tedious than checking the rank condition (32).

B. Improved VI-Based Algorithm

It can be seen that the VI method boils down to solving various sequences of linear equations, whose computational cost may be formidable for a large MAS. Thus, it is interesting to further improve our data-driven VI algorithm over (31) as follows.

1) In (29), letting $P_k = I_{n+n_z}$ and noting $K_k = -J^T$ give

$$\begin{aligned} &|\hat{\xi}_i(t+\delta t)|_{I_{n+n_z}} - |\hat{\xi}_i(t)|_{I_{n+n_z}} \\ &= \int_t^{t+\delta t} [|\hat{\xi}_i|_{Y^T+Y} + 2u_i^T J^T \hat{\xi}_i + 2\eta_i^T \begin{bmatrix} E_i \\ 0 \end{bmatrix}^T \hat{\xi}_i \\ &\quad + 2\hat{\gamma}_i^T \hat{\xi}_i] d\tau \\ &= \int_t^{t+\delta t} [|\hat{\xi}_i|_{Y^T+Y} + 2u_i^T B^T x_i + 2\eta_i^T E_i^T x_i \\ &\quad + 2\hat{\gamma}_i^T \hat{\xi}_i] d\tau. \end{aligned} \quad (36)$$

Equation (36) together with (30) leads to

$$\hat{\Psi}_0 \begin{bmatrix} \text{vecs}(Y^T + Y) \\ \text{vec}(B^T) \\ \text{vec}(E_i^T) \end{bmatrix} = \hat{\Phi}_0 \quad (37)$$

where

$$\hat{\Psi}_0 = [\Gamma_{\hat{\xi}_i \hat{\xi}_i} M, 2\Gamma_{\hat{\xi}_i u_i}, 2\Gamma_{\hat{\xi}_i \eta_i}]$$

$$\hat{\Phi}_0 = (\delta_{\hat{\xi}_i} - 2\Gamma_{\hat{\xi}_i \hat{\gamma}_i} M) \text{vecs}(I_{n+n_z}).$$

Solving (37) gives B and E_i , and hence J and $\begin{bmatrix} E_i \\ 0 \end{bmatrix}$.

2) With J and $\begin{bmatrix} E_i \\ 0 \end{bmatrix}$ available, (31) reduces to

$$\hat{\Psi}'_i [\text{vecs}(H_k)] = \hat{\Phi}'_{ik} \quad (38)$$

where

$$\begin{aligned} \hat{\Psi}'_i &= \Gamma_{\hat{\xi}_i \hat{\xi}_i} M \\ \hat{\Phi}'_{ik} &= \begin{bmatrix} \delta_{\hat{\xi}_i} - 2\Gamma_{\hat{\xi}_i \hat{\gamma}_i} M - 2\Gamma_{\hat{\xi}_i u_i} (I_{n+n_z} \otimes J^T) M \\ -2\Gamma_{\hat{\xi}_i \eta_i} \left(I_{n+n_z} \otimes \begin{bmatrix} E_i \\ 0 \end{bmatrix}^T \right) M \end{bmatrix} \text{vecs}(P_k). \end{aligned}$$

In this way, H_k can be iteratively solved from (38).

As in Algorithm 1, $\tilde{P}_{k+1}$ can be expressed as $\tilde{P}_{k+1} \triangleq P_k + \epsilon_k (H_k - P_k J J^T P_k + I_{n+n_z})$. Thus, by iteratively applying the solution of (38) to Algorithm 1, one can obtain the approximate solution to (8) and the feedback control gain (21) without using the information of system matrices.

We further establish the solvability of (37) and (38) by the following two lemmas.

Lemma 4: The matrix $\hat{\Psi}_0$ in (37) has full column rank if

$$\begin{aligned} \text{rank}([\Gamma_{\hat{\xi}_i \hat{\xi}_i}, \Gamma_{x_i u_i}, \Gamma_{x_i \eta_i}]) &= \frac{(n + n_z)(n + n_z + 1)}{2} \\ &+ n(m + q). \end{aligned} \quad (39)$$

Proof: The proof of this lemma is similar to that of Lemma 3 and is thus omitted.

Lemma 5: The matrix $\hat{\Psi}'_i$ in (38) has full column rank if

$$\text{rank}([\Gamma_{\hat{\xi}_i \hat{\xi}_i}]) = \frac{(n + n_z)(n + n_z + 1)}{2}. \quad (40)$$

Proof: The proof of this lemma is similar to that of Lemma 3 and is thus omitted.

Remark 6: Since (38) is solvable if the rank condition (39) is satisfied, (39) is the only rank requirement for our improved VI-based algorithm. The rank condition (39) is milder than (32) as the column dimension of the matrix to be tested is reduced by $n_z(m + q)$.

Remark 7: Since, in our improved VI-based algorithm, H_k is iteratively solved from (38), the number of the unknown variables of (38) is reduced by $(n + n_z)(m + q)$ compared with (31). Thus, our improvements reduce the computational cost more profoundly than using (31).

Remark 8: Another merit of our VI algorithm is that the unknown variables of (38) are independent of the index i . By taking advantage of the connectivity of the graph, there is no need to solve (38) for all $i = 1, \dots, N$. For example, let $\mathcal{G}^- = \{\mathcal{V}^-, \mathcal{E}^-\}$ be a subgraph of $\mathcal{G}$ with $\mathcal{V}^- = \{1, \dots, N\}$, and $\mathcal{E}^-$ being obtained from $\mathcal{E}$ by removing all nodes incident to node 0. Suppose that $\mathcal{G}^-$ contains a spanning tree with one of the children of $\mathcal{V}^-$ as its root. Then, it suffices to solve (38) with respect to some index $1 \leq i \leq N$ corresponding to the root of $\mathcal{G}^-$. Then, the solution of (38) can be passed to other followers by the graph $\mathcal{G}^-$.

Our improved VI-based algorithm is summarized as Algorithm 2.

To obtain a concrete comparison with the VI method in [18], consider an example with $n = 10, m = 8, p = 4, q = 20, n_z = 40$, and $N = 5$. Assume that each follower has 2 neighbors, i.e., $|\mathcal{N}_i^-| = 2$ and $h_i = 2(n + n_z) = 100$. Tables I and II show, for each follower, the numbers of unknown variables and rank conditions of the two VI-based algorithms, respectively. The contrast is stark.

Algorithm 2: Improved VI-Based Algorithm for CORP.

```

1:  $i = 1$ 
2: repeat
3:   Choose  $P_0 = P_0^T > 0$ .  $k, q \leftarrow 0$ . Apply any locally
     essentially bounded initial input  $u_i^0$ . Choose a proper  $t_0$  such
     that  $|\rho_i|$  is small enough. Collecting data from  $t_0$  until the
     rank condition (39) is satisfied.
4:   Solve  $B$  and  $E_i$  from (37). Obtain  $J$  and  $\begin{bmatrix} E_i \\ 0 \end{bmatrix}$ .
5:   loop
6:     Solve  $H_k$  from (38)
7:      $\tilde{P}_{k+1} \leftarrow P_k + \epsilon_k (H_k - P_k J J^T P_k + I_{n+n_z})$ 
8:     if  $\tilde{P}_{k+1} \notin B_q$  then
9:        $P_{k+1} \leftarrow P_0$ .  $q \leftarrow q + 1$ .
10:    else if  $|\tilde{P}_{k+1} - P_k|/\epsilon_k < \varepsilon$  then
11:      return  $K_k = -J^T P_k$ 
12:    else
13:       $P_{k+1} \leftarrow \tilde{P}_{k+1}$ 
14:    end if
15:     $k \leftarrow k + 1$ 
16:  end loop
17:   $K^* = [K_x^*, K_z^*] = \omega^{-1} K_k$  with
      $0 < \omega \leq 2\text{Re}(\lambda_i), \forall \lambda_i \in \sigma(DH), i = 1, \dots, N$ .
18:  Obtain the following controller

19:   $i = i + 1$ 
20: until  $i = N + 1$ 

```

TABLE I
COMPARISON OF COMPUTATIONAL COMPLEXITY

Algorithm	Equation Number	Unknown variables
[18]'s VI Algorithm	(34)	7675
Our VI Algorithm	(38)	1275

TABLE II
COMPARISON OF RANK CONDITION

Algorithm	Rank Condition for Data Collecting
[18]'s VI Algorithm	$\text{rank}([\Gamma_{\hat{\xi}_i \hat{\xi}_i}, \Gamma_{\hat{\xi}_i u_i}, \Gamma_{\hat{\xi}_i \eta_i}, \Gamma_{\hat{\xi}_i \psi_i}]) = 7675$
Our VI Algorithm	$\text{rank}([\Gamma_{\hat{\xi}_i \hat{\xi}_i}, \Gamma_{x_i u_i}, \Gamma_{x_i \eta_i}]) = 1555$

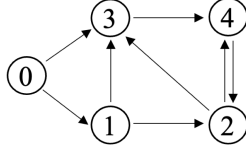
IV. APPLICATION

In this section, we use an example to illustrate the effectiveness of our improved VI-based algorithm.

Let's consider a MAS with four followers as follows:

$$\begin{aligned} A &= \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, E_i = \begin{bmatrix} 0 & 0 \\ 0 & 1.5i \end{bmatrix} \\ C &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, F = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, S = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \end{aligned} \quad (42)$$

where $i = 1, 2, 3, 4$. It can be easily verified that Assumptions 1–3 are all satisfied. By (4), the minimal 2 – copy internal model of S is as

Fig. 1. Communication digraph $\mathcal{G}$.

follows:

$$G_1 = \text{blockdiag}(S, S) \text{ and } G_2 = \text{blockdiag}\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right). \quad (43)$$

The communication digraph is shown in Fig. 1. It can be observed that the digraph $\mathcal{G}$ satisfies Assumption 4.

The matrices $\mathcal{A}$ and H of the digraph $\mathcal{G}$ are given by

$$\mathcal{A} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}, H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 2 & 0 & -1 \\ -1 & -1 & 3 & 0 \\ 0 & -1 & -1 & 2 \end{bmatrix}.$$

Thus, the matrix $D = \text{blockdiag}(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{2})$. The eigenvalues of DH are 0.380, 1.000, $1.310 \pm 0.196i$.

Remark 9: It is noted that the dimension p of the output for this example is 2, and the communication digraph contains loops and is thus not acyclic. Therefore, the approach in [18] does not work for this example.

The improved VI-based approach in Section III is then applied to solve the CORP for (42). Let $\omega = 1.1\min(\text{Re}(\lambda_i))$, where $\lambda_i \in \sigma(DH)$, $i = 1, \dots, 4$. By (21), we obtain the exact feedback gain $K = [K_x, K_z] = -\omega^{-1}J^T P^*$ as follows:

$$K = \begin{bmatrix} -0.605 & -7.796 & 0.305 & -0.600 & 0.934 & -3.181 \\ -2.729 & -0.605 & 0.988 & -3.165 & -0.079 & 0.668 \end{bmatrix}. \quad (44)$$

Set the initial input $u_i^0 = K_0' \hat{\xi}_i + \delta_i$ with the initial feedback gain K_0' and exploration noise δ_i as follows:

$$K_0' = \begin{bmatrix} -3 & -3 & 1 & 0 & 0 & -2 \\ -2 & -3 & 2 & -3 & 1 & 1 \end{bmatrix}$$

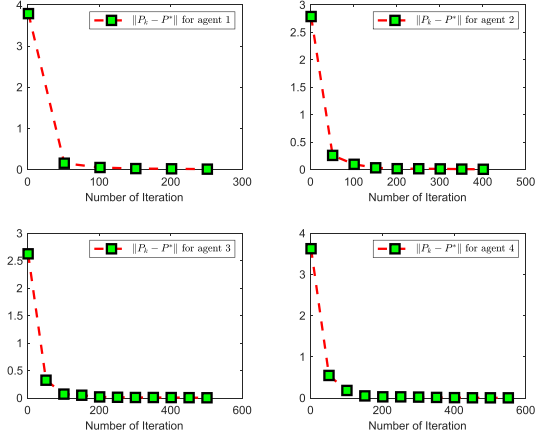
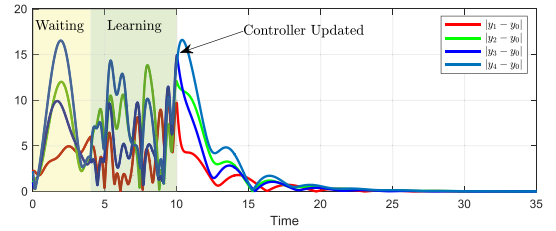
$\delta_i = 10$

$$\begin{bmatrix} \sin(-4t) + \sin(-3t) + \sin(-t) + \sin(2t) + \sin(5t) + \sin(7t) \\ \sin(-7t) + \sin(-5t) + \sin(-3t) + \sin(2t) + \sin(4t) + \sin(8t) \end{bmatrix},$$

and set $\mu = 15$ for (23). Let $t_j = 4 + 0.1j$ s for $j = 0, 1, \dots, s$ with $s = 60$. We start the iteration process with $P_0 = iI_6$ for $i = 1, 2, 3, 4$, and the sets B_q are defined as $B_q = \{Q \geq 0 \mid |Q| < 30(q+1)\}$ for $Q = Q^T \in \mathbb{R}^{6 \times 6}$, $q \in \mathbb{N}$. It is verified that the rank condition (39) is satisfied by using the data collected between [4, 10]s, and it takes 297, 411, 505, and 572 steps for agents 1–4 to meet the convergence criterion $|\bar{P}_{k+1} - P_k|/\epsilon_k < \epsilon$ with $\epsilon_k = 3/k$ and $\epsilon = 0.01$, respectively. The comparison of P_k and its target value P^* for each agent is shown in Fig. 2.

The learned feedback gains K_i^* for $i = 1, 2, 3, 4$ are as follows:

$$K_1^* = \begin{bmatrix} -0.606 & -7.794 & 0.307 & -0.600 & 0.932 & -3.179 \\ -2.727 & -0.606 & 0.986 & -3.164 & -0.079 & 0.665 \end{bmatrix}$$

Fig. 2. Comparison of P_k and their target values P^* for improved VI-based method.Fig. 3. Plots of the tracking error $|y_i - y_0|$ for improved VI-based method.

$$K_2^* = \begin{bmatrix} -0.606 & -7.795 & 0.307 & -0.600 & 0.933 & -3.179 \\ -2.728 & -0.606 & 0.986 & -3.164 & -0.079 & 0.665 \end{bmatrix}$$

$$K_3^* = \begin{bmatrix} -0.605 & -7.795 & 0.306 & -0.601 & 0.934 & -3.179 \\ -2.728 & -0.605 & 0.987 & -3.164 & -0.079 & 0.666 \end{bmatrix}$$

$$K_4^* = \begin{bmatrix} -0.605 & -7.795 & 0.305 & -0.601 & 0.935 & -3.180 \\ -2.728 & -0.605 & 0.987 & -3.164 & -0.079 & 0.667 \end{bmatrix}.$$

The controller in the form of (41) with the learned feedback gain starts to operate at $t = 10$ s. As shown in Fig. 3, our learned controller works satisfactorily.

Remark 10: It is noted that the rank condition (32) is not satisfied for $i = 2, 3, 4$ using the data between [4, 10] s. Thus, (31) does not work for this particular scenario.

Remark 11: Finally, it is noted that since node 1 is a root of the graph $\mathcal{G}^-$, by Remark 8, it suffices to solve (38) with $i = 1$ only.

V. CONCLUSION

In this article, we have investigated the CORP for unknown linear homogeneous MASs using data-driven distributed internal model approach. Compared with existing results in the literature, this article has offered three features. First, we have solved the problem for MASs whose followers are governed by MIMO systems and removed the assumption that the communication network is acyclic. Second, the sequence of equations (34) and the corresponding solvability condition (35) of [18] have been significantly simplified to (31) and (32), respectively. Third, we have further improved our VI algorithm by reducing (31) to lower-dimensional (38). A numerical example has been used to demonstrate the effectiveness of our improved VI approach. The future work will concentrate on the same problem for unknown linear heterogeneous MASs.

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